

BLAGOVESHCHENSK, Russian Federation

WMO: 315100

Lat: 50.280N

Lon: 127.478E

Elev: 169

StdP: 99.31

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.2	(c) -30.0	(d) -35.1	(e) 0.1	(f) -31.7	(g) -33.0	(h) 0.2	(i) -29.2	(j) 6.3	(k) -16.2	(l) 5.5	(m) -16.9	(n) 1.1	(o) 260	(p) 0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	31.2	21.1	29.5	20.6	27.9	20.0	23.5	28.1	22.5	27.0	21.6	25.9	2.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.1	26.1	21.0	16.0	24.9	20.1	15.1	23.9	71.3	28.1	67.4	27.0	64.1	26.1	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 5.4	(c) 4.7	(e) -33.7	(f) 34.3	(g) 2.9	(h) 1.8	(i) -35.8	(j) 35.6	(k) -37.5	(l) 36.7	(m) -39.1	(n) 37.7	(o) -41.3	(p) 39.0		
(a) 6.3	(b) 5.4	(c) 4.7	(e) -33.9	(f) 25.4	(g) 2.9	(h) 1.4	(i) -36.0	(j) 26.4	(k) -37.7	(l) 27.3	(m) -39.3	(n) 28.1	(o) -41.3	(p) 29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-20.5	-15.9	-6.4	4.9	13.2	19.7	22.5	20.2	13.4	3.8	-9.3	-19.3		
	DBStd	15.95	4.89	4.97	5.96	4.32	4.10	3.54	2.51	2.94	3.92	4.71	6.43	4.52		
	HDD10.0	4054	946	724	509	162	15	0	0	0	13	199	578	907		
	HDD18.3	6142	1204	957	768	403	168	26	2	15	154	451	828	1166		
	CDD10.0	1237	0	0	0	9	113	291	388	316	114	6	0	0		
	CDD18.3	282	0	0	0	0	8	66	132	72	5	0	0	0		
	CDH23.3	2204	0	0	0	8	135	601	948	458	53	1	0	0		
	CDH26.7	629	0	0	0	1	36	199	283	105	5	0	0	0		
	WSAvg	2.0	1.8	1.9	2.3	2.6	2.4	1.9	1.7	1.8	1.9	2.2	2.0	1.8		
	PrecAvg	559	6	5	11	30	47	84	134	123	70	24	13	10		
(15) Precipitation	PrecMax	809	36	20	38	139	143	191	269	252	181	84	40	24		
	PrecMin	357	0	0	0	3	5	6	24	0	11	0	0	2		
	PrecStd	95	6	5	9	24	30	47	61	56	41	19	11	5		
	PrecStd	95	6	5	9	24	30	47	61	56	41	19	11	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.7	1.2	11.5	24.1	30.3	34.0	33.9	31.4	27.2	20.1	8.8	-5.2		
		MCWB	-8.5	-2.7	3.9	11.6	15.5	20.8	23.0	22.3	18.3	10.8	3.5	-6.9		
	2%	DB	-9.5	-2.6	7.9	18.9	27.1	31.1	31.4	29.2	24.9	16.1	4.6	-8.1		
		MCWB	-11.0	-5.1	2.1	8.6	14.9	19.8	22.5	21.6	17.1	9.6	0.8	-9.4		
	5%	DB	-11.4	-5.6	4.9	15.7	24.1	28.9	29.6	27.7	22.7	13.2	1.7	-10.6		
		MCWB	-12.6	-7.6	0.2	7.0	13.2	19.2	21.7	20.6	15.9	7.5	-1.0	-11.8		
	10%	DB	-13.3	-8.2	2.5	13.1	21.4	26.8	28.0	26.0	20.3	10.9	-0.6	-12.7		
		MCWB	-14.3	-9.8	-1.2	5.7	12.2	18.5	21.1	20.1	14.4	6.0	-2.5	-13.6		
	0.4%	WB	-8.5	-2.4	4.3	11.8	17.2	23.4	25.1	24.2	20.1	12.9	3.6	-6.5		
		MCDB	-7.1	0.6	10.5	22.4	25.7	30.2	30.2	28.8	24.4	17.7	7.8	-5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-10.9	-5.0	2.3	9.2	15.8	21.6	23.9	22.9	18.3	10.1	1.0	-9.4		
		MCDB	-9.4	-2.8	7.1	17.4	24.4	28.1	28.5	26.8	22.7	14.7	4.2	-8.2		
	5%	WB	-12.6	-7.6	0.5	7.6	14.5	20.5	23.0	21.8	17.0	8.2	-0.7	-11.8		
		MCDB	-11.4	-5.7	4.4	14.5	22.1	26.5	27.5	26.0	21.2	12.4	1.5	-10.7		
	10%	WB	-14.3	-9.8	-1.0	6.1	13.3	19.4	22.2	20.8	15.6	6.5	-2.6	-13.6		
		MCDB	-13.3	-8.2	2.3	12.3	19.8	25.1	26.4	24.6	19.0	10.5	-0.7	-12.7		
	5% DB	MDBR	9.5	11.3	10.7	11.0	11.5	10.5	9.0	9.2	10.5	9.3	8.3	8.3		
		MCDBR	10.7	13.3	12.6	16.4	16.5	14.2	11.8	11.8	14.1	13.6	10.6	9.4		
		MCWBR	9.4	11.0	8.1	8.8	6.8	5.7	4.8	5.2	6.9	7.8	7.4	8.2		
		MCWBR	10.6	13.3	11.7	14.8	14.0	11.8	9.6	9.7	11.3	12.3	10.2	9.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.4	11.1	7.7	8.4	6.4	5.6	4.8	5.1	6.3	7.7	7.3	8.2		
		MCWBR	9.4	11.1	7.7	8.4	6.4	5.6	4.8	5.1	6.3	7.7	7.3	8.2		
	Clear-Sky Solar Irradiance	taub	0.272	0.302	0.374	0.508	0.567	0.501	0.483	0.430	0.393	0.386	0.319	0.278		
		taud	2.161	2.090	1.954	1.788	1.731	2.005	2.109	2.242	2.264	2.118	2.150	2.118		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	750	815	806	731	707	768	776	802	791	700	670	667		
		Edh at Noon	76	107	148	198	219	168	149	125	109	105	77	69		
(44) All-Sky Solar Radiation	RadStd	RadAvg	1.67	2.88	4.41	5.00	5.44	5.85	5.42	4.81	3.87	2.57	1.65	1.30		
		RadStd	0.06	0.10	0.28	0.53	0.45	0.74	0.46	0.34	0.33	0.15	0.10	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page